

Total No. of Questions : 10]

SEAT No. :

P-9349

[Total No. of Pages : 2

[6181]-584

B.E. (Mechanical Engineering)
ADVANCED MANUFACTURING PROCESSES
(2015 Pattern) (Semester - II) (402050A) (Elective - IV)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.

Q1) a) Explain with neat sketch electro hydraulic forming. Write its applications. **[6]**

b) Write down four points of comparison between Flow forming and Spinning Process. **[4]**

OR

Q2) a) Explain with neat sketch Incremental sheet metal forming and list their applications. **[6]**

b) Explain the construction and Working of Laser Beam welding. **[4]**

Q3) a) Explain with neat sketch Electro Jet Machining process with its process parameters. **[6]**

b) Write short note on Cryogenic welding. **[4]**

OR

Q4) a) Explain principle, working & applications of 'Magnetic Abrasive Finishing' with neat line diagram. **[6]**

b) Explain the principal of laser based heat treatment and name any four different types of laser sources used in heat treatment. **[4]**

Q5) a) Describe single point diamond turning in terms of its process characteristics, the machine tool, material machined from quality control aspects. **[6]**

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- b) Explain the process of Focused Ion Beam Machining. [6]
- c) Explain the need of micro and nano machining techniques. [4]

OR

- Q6)** a) Explain Ultrasonic Micromachining process in detail with applications. [6]
- b) Explain concept of Photo Lithography with neat sketch. Write its applications. [6]
 - c) What are the challenges in micro and nano machining techniques? [4]

- Q7)** a) What is additive manufacturing process? Describe the different steps by which a part or component is build using additive manufacturing process. [8]
- b) State the working principal of Powder Based Fusion Additive Manufacturing Process with neat sketch. Write down its advantages, limitations and applications (2 Each). [8]

OR

- Q8)** a) Explain the working principle and process steps of extrusion based (FDM) additive manufacturing process with neat sketch. [8]
- b) i. Write a short note on 'Design for Additive Manufacturing'. [4]
 - ii. Write applications of 'Additive Manufacturing' in Medical & Aerospace industries. [4]

- Q9)** a) Explain operating principle of 'Atomic Force Microscope' with neat sketch. [6]
- b) Explain operating principle of 'Scanning Tunneling Microscope' with neat sketch. [6]
 - c) Explain with sketch operating principle of 'Nuclear Magnetic Resonance Spectroscopy'. [6]

OR

- Q10)** a) Explain operating principle of 'Scanning Electron Microscopes' with neat sketch. [6]
- b) Explain operating principle of 'Transmission Electron Microscope' with neat sketch. [6]
 - c) Explain with sketch operating principle of 'Energy-dispersive X-ray spectroscopy'. [6]

